Dr. Tooba Quidwai

Molecular & Cell Biologist | Advanced Imaging & Cilia Biology | Translational Disease Models

Nationality: Indian
Email: tquidwaiunipune@gmail.com
tooba.quidwai@bio.ku.dk
Key Links: Google Scholar LinkedIn Twitter/X BlueSky

• Research Profile

I am a biomedical scientist with over a decade of international research experience across India, the UK, Germany, and the USA. My expertise lies in ciliary biology, vesicular trafficking, proteomics, genomics and advanced imaging (super-resolution microscopy, CLEM, 3D-TEM, STORM, SIM, FRAP, FRET). I acquired advanced imaging training before Ph.D. in the lab of Prof. Jonas Ries at EMBL Heidelberg, Germany.

During my PhD at the University of Edinburgh (Prof. Pleasantine Mill), I discovered a novel vesicular trafficking role for WDR35/IFT121, integrating 3D-TEM and correlative microscopy, mouse genetics, and proteomics to resolve coatomer-like functions at the cilium. As a postdoctoral researcher at UMass Chan Medical School (Prof. George Witman, and Dr. Sumeda Nandadasa), I advanced my skills in mouse models and Chlamydomonas biology, uncovering roles for ADAMTS9 and TMEM67 in kidney cyst formation and central apparatus role in flagellum function respectively.

My career highlights include 4 peer-reviewed publications including first and contributing author (2 in *eLife*, *eLIfe*, 1 in *AJHG*, 1 in *Chemical Science*), and one preprint, competitive fellowships (EMBO, MRC, ESRIC, Howard Hughes & Helmsley and Morey endorsement), and invited talks at international conferences (ASCB 2022, EMBO 2018).

• Education

1. **PhD, Genetics & Molecular Medicine** (15/09/2015 – 16/06/2020)
   MRC Institute of Genetics & Cancer, University of Edinburgh, UK
   Supervisor: Prof. Pleasantine Mill
   Thesis: The role of WDR35 in the formation of functional cilia
2. **MSc, Biotechnology** (01/06/2008 – 01/05/2010)
   Department of Biotechnology, University of Pune, India
   Grade: 4.15/6
3. **BSc, Botany, Zoology, Chemistry** (01/06/2004 – 01/07/2007)
   Dr. Ram Manohar Lohia Avadh University, India
   Grade: First Class

• Current Status

1. **Research Contributions:** Contributing remotely to international research grants, including Danish DFF1-2025, and Marie Skłodowska-Curie Actions (MSCA). These projects, in collaboration with scientists at the University of Copenhagen focus on obesity, kidney disease, and metabolic disorders. **Position at the University of Copenhagen was dependent on the success of above-mentioned grants. Secured seal of excellence certificate in MSCA-2026 Postdoctoral fellowship application.**

2. **Public Outreach:** Actively engaging in public and scientific outreach with Indian scientific institutions through seminars and interdisciplinary dialogue. So far covered (1) NIO-India, (2) BITS-India and more lined up at institutes and industries in Bangalore, India.
3. **Upskilling:** Pursuing online certified courses (see list on last page) to expand expertise in computational biology, imaging, science communication and disease mechanism topics.

• Research Experience

1. **Postdoctoral Research Associate** – (01/08/2021 – 01/09/2023)
   Employer - UMass Chan Medical School, Worcester, MA, USA
   Supervisors: Prof. George B. Witman, Dr. Sumeda Nandadasa
   Focus: Ciliary biology in *Chlamydomonas* and mouse models to explore role of central apparatus and role of ADAMTS9-mediated TMEM67 cleavage in kidney cystogenesis respectively
   Techniques: Mouse genetics, kidney histology, 3D-whole kidney imaging, flagellar preparation, Immunoprecipitation (IP) and mass spectrometry of *Chlamydomonas* flagellar central apparatus proteins and IP of ADAMTS-9 pathway proteins, Bioinformatics and structural modelling of TMEM67

2. **Bridging Postdoctoral Fellow** – (20/12/2019 – 31/07/2021)
   Employer - MRC Institute of Genetics & Molecular Medicine, University of Edinburgh, UK
   Supervisor - Prof. Pleasantine Mill, Prof. Yanick Crow
   Focus- EM and CLEM to study WDR35 function and nuclear cGAS localization defects
   Techniques – 3D-TEM, CLEM (correlative light and electron microscopy), confocal, airy scan microscopy

3. **PhD Researcher** – (15/09/2015 – 19/12/2019)
   Employer - MRC Institute of Genetics & Molecular Medicine, University of Edinburgh, UK
   Supervisor- Prof. Pleasantine Mill
   Techniques – Imaging (confocal, STED, SIM, Hyvolution, deconvolution, airy scan, spinning disk, transmission light-DIC, phase contrast), mouse genetics, fibroblast and cell line generation, genome engineering including CRISPR, Proteomics including immunoprecipitation and mass spec, bioinformatics including structural modeling. Electron microscopy including 3D-TEM, CLEM and immunogold labeling, ImageJ, Imaris, image processing.
   Key achievement - Identified coatomer-like vesicle transport function of WDR35/IFT121 *(eLife 2021)*. contributed to PCM-1 *(eLife 2023)* and PLAA (*American Journal of Human Genetics* projects 2017).

4. **Visiting Scientist** – (01/04/2014 – 14/09/2015)
   Employer - European Molecular Biology Laboratory (EMBL) Heidelberg, Germany
   Supervisor - Prof. Jonas Ries
   Techniques – Optimization of buffers, dyes (photoactivable/ convertible/caged) and protein tags (SNAP/Halo) for STORM/PALM imaging. 3D and dual color PALM imaging of Platelet cytoskeleton, MATLAB, ImageJ.
   Key achievements – Published normal and activated cytoskeleton of platelets *(BioRxiv 2017)* and novel caged dyes for live-cell PALM (*Chemical Science 2017*)

5. **Visiting Scientist-** (01/06/2012 – 31/03/2014)
   Employer- Max Planck Institute of Molecular Physiology, Dortmund, Germany
   Supervisor - Prof. Philippe Bastiaens
   Focus- The Ras pathway, signal transduction pathway in cilia and cell profiling of endogenous levels of H/N/K-Ras and PDEδ proteins in hepatic carcinoma HepG2 and MDCK cells in response to anti-cancer drugs. Ras pathway in cilia, protein–protein interactions via in-situ FRET, RNAi-based ciliation assays.
   Techniques – Imaging, stable cell lines, RNAi, cell proliferation assays, image processing

6. **Junior Research Fellow** – (20/09/2010 – 10/05/2012)
   Employer - National Institute of Immunology, India
   Supervisor - Dr. Apurba K. Sau
   Focus - Functional domains study of IFNγ induced GTPase human guanylate binding protein (hGBP1)
   Techniques – Site directed mutagenesis, microbiology, bioinformatics including structural modeling, and biochemistry and proteomics methods including various protein purification (HPLC, and affinity chromatography), Circular dichroism (CD) and radioactive isotope-based GTPase assay to study GTPase function of hGBP-1 protein
   Key achievement: Acknowledgement: *FEBS journal 2020*

• Publications

1. Emma A Hall, Dhivya Kumar, Suzanna L Prosser, Patricia L Yeyati, Vicente Herranz-Pérez4, Jose Manuel García-Verdugo, Lorraine Rose, Lisa McKie, Daniel O Dodd, Peter A Tennant, Roly Megaw, Laura C Murphy, Marisa F Ferreira, Graeme Grimes, Lucy Williams, ***Tooba Quidwai***, Laurence Pelletier, Jeremy F Reiter, Pleasantine Mill. 2023. Centriolar satellites expedite mother centriole remodeling to promote ciliogenesis. *eLIFE*. https://doi.org/10.7554/eLife.79299
2. ***Tooba Quidwai***, Jiaolong Wang, Emma A. Hall, Narcis Adrian Petriman, Weihua Leng, Petra Kiesel, Jonathan N. Wells, Laura C. Murphy, Margaret A. Keighren, Joseph A. Marsh, Esben Lorentzen, Gaia Pigino, Pleasantine Mill. 2021. A WDR35-dependent coat protein complex transports ciliary membrane cargo vesicles to cilia. *eLIFE*. https://doi.org/10.7554/eLife.69786
3. Aastha Mathur, Sandra Raquel Correia, Serge Dmitrieff, Romain Gibeaux, Iana Kalinina, ***Tooba Quidwai***, Jonas Ries, Francois Nedelec. 2018. Cytoskeleton mechanics determine resting size and activation dynamics of platelets. *BioRxiv*. https://doi.org/10.1101/413377. (*Was in review in eLIFE*).
4. Emma A. Hall, Michael S. Nahorski, Lyndsay M. Murray, Ranad Shaheen, Emma Perkins, Kosala N. Dissanayake, Yosua Kristaryanto, Ross A. Jones, Julie Vogt, Manon Rivagorda, Mark T. Handley, Girish R. Mali, ***Tooba Quidwai***, Dinesh C. Soares, Margaret A. Keighren, Lisa McKie, Richard L. Mort, Noor Gammoh, Amaya Garcia-Munoz, Tracey Davey, Matthieu Vermeren, Diana Walsh, Peter Budd, Irene A. Aligianis, Eissa Faqeih, Alan J. Quigley, Ian J. Jackson, Yogesh Kulathu, Mandy Jackson, Richard R. Ribchester, Alex von Kriegsheim, Fowzan S. Alkuraya, C. Geoffrey Woods, Eamonn R. Maher, Pleasantine Mill. 2017. *PLAA* Mutations Cause a Lethal Infantile Epileptic Encephalopathy by Disrupting Ubiquitin-Mediated Endolysosomal Degradation of Synaptic Proteins. *American Journal of Human Genetics* **100**:P706-P724. https://doi.org/10.1016/j.ajhg.2017.03.008
5. Sebastian Hauke, Alexander Von Appen, ***Tooba Quidwai***, Jonas Ries, Richard Wombacher. 2017. Specific protein labeling with caged fluorophores for dual-color imaging and super-resolution microscopy in living cells. *Chemical Science* **8:**559-566. https://doi.org/10.1039/C6SC02088G
6. Acknowledgement for proteomics work (Protein purification, CD, biochemical assays): Stimulation of GMP formation in hGBP1 is mediated by W79 and its effect on the antiviral activity. *The FEBS journal* 2020. https://doi.org/10.1111/febs.15611

H-index: 9 (Google Scholar, 207 citations)

• Fellowships and Awards

1. **MCSA postdoctoral fellowship Seal of Excellence certificate (2026):** Secured in collaboration with Prof. Lotte Pedersen lab and Københavns University University on the topic: Role of CCDC198/FAME in ciliary biogenesis and function: implications for kidney disease, obesity and metabolic disorders

2. **EMBO Short-Term Fellowship, 2018**: Supporting EM and CLEM imaging at MPI-CBG Dresden, Germany in the lab of Prof. Gaia Pigino from 01/10/2018 to 30/12/2018
3. **Howard Hughes & Helmsley Travel Awards (2016):** Supporting microscopy training at quantitative imaging course from 05/04/2016-18/04/2016 at Cold Spring Harbor Laboratory, USA
4. **Three-year ESRIC & MRC Ph.D. Fellowship (2015–2018):** Supporting Ph.D. research at Institute of Genetics and Cancer, University of Edinburgh, UK in the lab of Dr. Pleasantine Mill from 14/09/2015 to 14/09/2018
5. **Moray Endorsement Fund, Edinburgh (2017)** – Short fellowship supporting my Ph.D. work in the lab of Prof. Pleasantine Mill and Genetics and Cancer, University of Edinburgh, UK

• Podium Presentations (Invited/Selected)

1. **National Institute of Oceanography, India (31/07/2025) Foundation Day Diamond Jubilee Lecture-** Invited talk
   Topic: Visualizing Cellular Machines: Insights into Cilia Biology and Translational Imaging
2. **BITS Pilani, Goa, India (30/07/2025) –** Invited talk
   Topic: Prospects of ciliary biology courses
   Visualizing Cellular Machines: Insights into Cilia Biology and Translational Imaging
3. **ASCB Cell Biology Conference, Washington DC (07/12/2022)** – Selected talk
   Topic: A WDR35/IFT121-dependent vesicular transport to cilia
4. **International Virtual Membrane Trafficking Seminar (06/05/2021)** – Invited talk
   Topic: Role of WDR35 in vesicular transport to cilia.
5. **BSCB GenSoc UK Cilia Spring Meeting, Edinburgh, UK (28/04/2020)** – Selected talk
   Topic: WDR35/IFT121-dependent coatomer transports ciliary membrane proteins from the Golgi to the cilia
6. **Celebrating BAME Women in Science, Edinburgh (07/10/2020)** – Invited outreach talk
   Topic: My experiences with Microscopes and Biology
7. **Edinburgh Super-resolution Imaging Consortium Symposium, Edinburgh, UK (23/06/2018)** – Invited talk
   Topic: Imaging methods to study cilia dynamics
8. **Scottish Microscopy Conference, Edinburgh, UK (24/11/2016)** – (Selected talk)
   Topic: WDR35/IFT121 Regulates Entry of Membrane Proteins to Cilia

• Courses attended

1. **ESRIC Super-Resolution Summer School (2016):** University of Edinburgh and Heriot-Watt University, Edinburgh, UK (08/08/2016 –12/08/2016)
2. **Cold Spring Harbor Laboratory Quantitative Imaging: From Cells to Molecules Course (2016):** Cold Spring Harbor, New York, USA (05/04/2016 – 18/04/2016)
3. **EMBL Software Carpentry Course (2015)** Topics covered included Unix shell, Python scripting, version control with GIT and GitHub, Python scientific stack, unit testing with Python, cluster resource usage, and basics of web development. EMBL, Heidelberg, Germany (04/11/2015 – 06/11/2015)

• Poster presentation

1. **EMBO Workshop, In Situ Structural Biology: Expanding the Toolbox for Structural Cell Biology 2025**
   Topic: Introducing a novel coatomer-like function of transport proteins (IFT-A) in the membrane protein trafficking to cilia. EMBL, Heidelberg, Germany, 4/02/2025 –07/02/2025
2. **Represented Prof. Pleasantine Mill's lab and the Medical Research Council (MRC) Disease Mechanism Section for the Quarterly Quinquennial Report 2017**
   Topic: The role of WDR35 in membrane protein transport and imaging methods to enhance the spatiotemporal resolution of cilia. September 2017

3. **Focus on Microscopy (FOM) Conference 2015**
   Topic: Platelet cytoskeleton imaged by STORM microscopy. Göttingen, Germany, 29/03/2015 – 01/04/2015.
4. **Labelling & Nanoscopy Conference, 2014**
   Topic: Buffers for super-resolution STORM microscopy. German Cancer Research Center (DKFZ), Heidelberg, Germany, 24/09/2014 –26/09/2014.
5. **Annual Meeting of the Indian Biophysical Society 2012.**
   Topic: Identification of a novel dimeric interface in full-length human guanylate binding protein. Chennai, India, 12/01/2012 –19/01/2012.

• Teaching, & Transferable Skills

1. **Trained peers in EM and advanced imaging 2021**. Trained senior researchers in Prof. Yanick Crow lab in electron microscopy techniques.
2. **Instructor at EMBO Single Molecule Course (2015)**. Assisting Prof. Jonas Ries in teaching the STORM super-resolution microscopy technique, at EMBL, Heidelberg, Germany
3. **Participated in organization of PCD patient outreach** events at University of Edinburgh, 2018.
4. **Master's Project at National Centre for Cell Science (NCC**S), India (Sept 2009- April 2010)
   Title: To study the effect of antiapoptotic agents (caspase and calpain inhibitors) on cryopreservation of hematopoietic stem cells with the help of FACS and confocal imaging with Dr. L.S. Limaye
5. **Master's project at Central Drug Research Institute (CDRI), India (June 2009- July 2009)**
   Title: Cloning, characterization, and protein purification of Apicoplast targeted protein in *Plasmodium falciparum w*ith Prof. Saman Habib

• Public Outreach

1. Contributed to public-facing content on ciliopathies 2018 (Cilia Lab, University of Edinburgh).
2. Speaker at *Celebrating BAME Women in Science* (Edinburgh, 2020).
3. Currently sharing my research journey and looking forward to learning from Indian scientific institutions through seminars, interdisciplinary dialogue spreading awareness about ciliopathies and related topics. So far covered (1) NIO-Goa-India-31/07/2025, (2) BITS-Pilani-Goa-30/07/2025 and a few upcoming in 2026).

• Technical Skills

1. **Imaging:** Confocal, brightfield fluorescence, transmission light microscopy methods, Light-based super-resolution imaging methods PALM/ STORM microscopy (dual-color and 3D), STED, SIM, Airy Scan, Hyvolution, Deconvolution, well versed with buffers/dyes/fluorophores/ and tags used for super-resolution microscope, Electron microscopy, CLEM and well trained with EM software like IMOD, tomography, and segmentation. FRET (fluorescence resonance energy transfer) and FRAP (fluorescence recovery after photobleaching)
2. **Bioinformatics:** Familiar with all common bioinformatics tools & structural modelling tools. Alpha fold, Swiss Model, Phyre^2, HHBLITS, Pymol, Coot and other standard software
3. **Software:** ImageJ, Imaris, Huygens, Adobe software — Photoshop, Illustrator, InDesign, Microsoft Word, Excel, and other standard software. MATLAB and Python (basic proficiency)
4. **Molecular Biology:** Isolation of plasmid & genomic DNA, PCR, RT-PCR, southern and northern blotting, cloning of gene constructs into different expression and transfection vectors, and colony hybridization using dig labelled probe & site-directed mutagenesis, and genome editing using CRISPR in mammalian cells
5. **Proteomics:** Immunoprecipitation (endogenous, GFP trap), mass spectrometry, protein isolation from various cell types, western blotting, protein quantification, SDS-PAGE, hands-on experience with different protein expression systems, and protein purification techniques like affinity, ion exchange, size exclusion

chromatography & HPLC, lyophilisation, ELISA, pulse-chase experiment using radioactive isotope and circular dichroism

6. **Animal cell culture:** Extensively handled in vitro cell culture of hematopoietic stem cells, MNCs from cord blood cells, and P. *falciparum* culture, making stable cell lines, transient transfection, and electroporation. Used various cell lines like MEF (mouse embryonic fibroblast), human fibroblast, RP2, IMCD3, MDCK, hepG2, hematopoietic stem cells, platelets, and *chlamydomonas* culture. FACS (fluorescence-activated cell sorting), MACS (magnetic-activated cell sorting)
7. **Mouse work:** Dissection and preparation of mouse embryonic fibroblast cells, genotyping, and breeding. Imaging Cryo and paraffin preserved mouse kidney tissue
8. **Plant tissue culture:** Nodal culture, somatic embryogenesis and in vivo & ex-vivo culture
9. **Microbiology:** Transformation, maxi/miniprep, and maintaining yeast culture
10. **Biochemical Assays:** Coupled enzyme ATPase & GTPase assays with radiolabeled isotopes and separating them on TLC (thin layer chromatography) and enzyme immobilization

• Strengths: International Mobility & Collaborative Track Record

As summarized above, I have extensive experience working and collaborating in diverse international environments, including India, Germany, the United Kingdom, and the United States. My PhD project (*eLife 2021*) was itself a highly collaborative effort between MRC-IGC (UK), MPI-Dresden (Germany), and Aarhus University (Denmark), requiring multiple research visits across UK and Germany to complete the experiments. These experiences, combined with my diverse technical skillset will help me adapt and successfully deliver innovative research projects

This proven record of mobility, collaboration, and interdisciplinarity aligns directly with success in challenging and collaborative projects, knowledge transfer, and researcher independence.

• Hobbies

Passionate about yoga, boxing, and volunteering to teach underprivileged children and promote health awareness in the community alongside medical professionals.

• Online Certified Courses

1. **Global Diabetes Challenge:** Copenhagen University, Coursera (Completed)
2. **Introduction to Programming with MATLAB** – Vanderbilt University, Coursera (Completed)
3. **Introduction to Data, Signal, and Image Analysis with MATLAB** – Vanderbilt University, Coursera (Completed)
4. **Programming for Everybody: Getting Started with Python** – University of Michigan, Coursera (Completed)
5. **Applied Bioinformatics Specialization (COVID-19 Analysis)** – UCSF, Coursera (Completed)
6. **Social Media Marketing in Practice Specialization** – Digital Marketing Institute, Coursera (Completed)
7. **Mastering Programming with MATLAB** – Vanderbilt University, Coursera (In Progress)
8. **Python for Everybody Specialization** – University of Michigan, Coursera (In Progress)
9. **Drug Hunting: The Science of Making New Medicines Specialization** – Novartis, Coursera (In Progress)
10. **Getting Started in Cryo-EM** – Grant J. Jensen, Coursera (Non-Certified)

• **References:** If needed